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PREMIUM NOTES

UPSC — Environment & Ecology Notes

UPSC Test Series — StuDy

High-yield facts, formulas and trap-radar — made for the real exam hall.

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UPSC — Environment & Ecology Notes

Ecology Fundamentals

- Ecosystem services (Millennium Assessment): provisioning, regulating, supporting, cultural. Ecological pyramid — energy pyramid is ALWAYS upright (10% law — Lindeman); number/biomass can invert (ocean biomass).
- Biogeochemical cycles: Carbon (photosynthesis/respiration), Nitrogen (fixation — Rhizobium, lightning; nitrification; denitrification), Phosphorus (no atmospheric phase — sedimentary, slowest), Sulphur, Water.
- Adaptations & interactions: mutualism (lichen — algae+fungi), commensalism (orchid on tree), parasitism, predation, competition; ecotone = transition zone (edge effect); ecological niche vs habitat.
- Succession: pioneer → climax; xerarch (rock), hydrarch (water); primary vs secondary (faster — soil exists). Invasive species in India: Lantana, Prosopis juliflora, water hyacinth (Eichhornia — 'terror of Bengal'), Parthenium.

Climate Change Architecture

- UNFCCC (1992 Rio) — principles: CBDR-RC; Kyoto (1997 — Annex I binding, CDM/JI/emissions trading); Paris (2015 — NDCs, 2°C/well below + 1.5°C pursuit, global stocktake every 5 yrs, ratchet mechanism).
- India's updated NDC: 45% emissions-intensity reduction by 2030 (from 2005), 50% cumulative electric installed capacity from non-fossil by 2030, net zero 2070 (announced COP26 Glasgow — Panchamrit: 500 GW non-fossil by 2030, one billion tonne reduction etc.).
- IPCC AR6 key lines: human-caused warming unequivocal; every increment worsens extremes; remaining carbon budget for 1.5°C shrinking. GHG inventory gases basket: CO_2, CH_4, N_2O, HFCs, PFCs, SF_6, NF_3.
- India bodies/schemes: NAPCC (2008 — 8 missions: solar, energy efficiency, water, Himalayan ecosystem, green India, sustainable habitat, agriculture, strategic knowledge), National Green Hydrogen Mission (2023), LiFE (Lifestyle for Environment — COP26 concept), International Solar Alliance (2015, HQ Gurugram — first treaty-based org headquartered in India).

Biodiversity Governance

- Biological Diversity Act 2002: NBA (Chennai) — approval for foreign access; State Biodiversity Boards; Biodiversity Management Committees (local — People's Biodiversity Registers). 2023 amendment: decriminalised offences, AYUSH exemptions debate.
- Wildlife Protection Act 1972: schedules rationalised by 2022 amendment (4 schedules now); CITES-aligned; NBWL chaired by PM; State Board by CM. Forest Conservation Act 1980 (post-2023: Van (Sanrakshan Evam Samvardhan) Adhiniyam — applicability clarifications for recorded forest areas).
- Forest cover data (ISFR — FSI, biennial): India forest+tree cover ~25% of geographical area (target 33% — National Forest Policy 1988); largest forest cover state — MP (area), highest % — Mizoram; mangrove cover ~5,000 km^2 (Sundarbans largest share). Assessment uses canopy density classes (VDF/MDF/OF).
- Keystone & indicator examples: tiger/sea otter (keystone), lichens (air pollution indicators — SO_2 sensitive), frogs (ecosystem health), vultures collapse (diclofenac — livestock drug; vulture restaurants & Jatayu centres as response).

Pollution, Waste & Energy Transition

- Air: primary pollutants (CO, SO_2, NOx, PM) vs secondary (ozone, PAN — photochemical smog). SAFAR/NAQI monitoring; GRAP (Graded Response Action Plan — Delhi NCR, CAQM statutory 2021).
- Water: BOD (biochemical oxygen demand — higher = more polluted), COD; dissolved oxygen sag; thermal pollution; groundwater contamination (arsenic — Bengal basin, fluoride — Rajasthan belt, nitrate — fertiliser overuse). CPCB/SPCB under Water Act 1974, Air Act 1981, EPA 1986 (umbrella — post Bhopal).
- EIA Notification 2006 (under EPA): Category A (central) / B (state); public hearing; TOR → EIA report → appraisal (EAC/SEAC) → clearance with conditions; draft EIA 2020 controversies (post-facto clearance, public consultation cuts).
- Energy mix targets: 500 GW non-fossil capacity by 2030; solar largest share; green hydrogen (electrolysers — renewable), ethanol blending E20 target (advanced to 2025–26), SATAT (CBG), PM-KUSUM (solar pumps), National Biofuel Policy. Battery/swapping & FAME→PM E-DRIVE transitions for EVs.

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